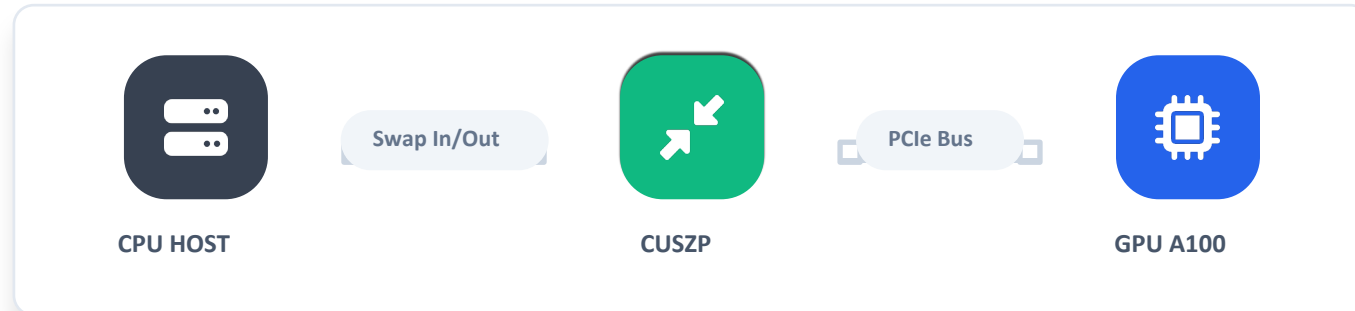


Compress-Transfer-Decompress for LLM Serving

cuSZp-Enabled CPU-GPU Data Pipeline in vLLM



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Presentation Agenda

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Motivation & Problem

Memory challenges & transfer bottlenecks in LLM serving

02

Proposed Solution

Compress-Transfer-Decompress methodology

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System Architecture

Integration of cuSZp with vLLM framework

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Implementation Details

H2D/D2H data flow and asynchronous execution

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Expected Results

Performance metrics, throughput, and accuracy analysis

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Status & Future Work

Current progress and roadmap for optimization

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Conclusion & Q&A

Summary of contributions and open discussion

The Memory Challenge in LLM Serving



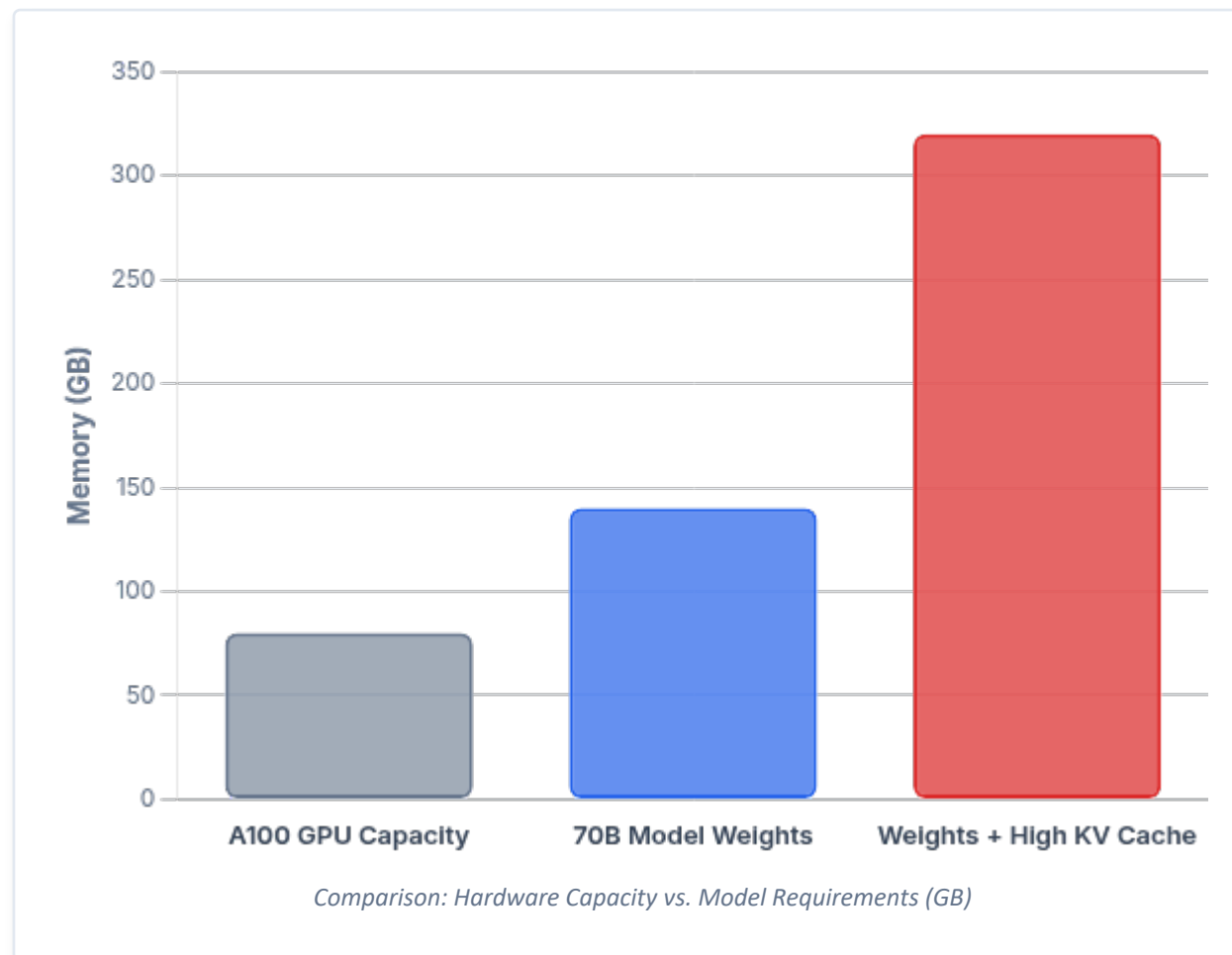
Large KV Cache Requirements

- LLMs must store Key-Value pairs for every token in the context window during inference.
- Size grows linearly with sequence length and batch size.
- Example: A 70B parameter model can require tens to hundreds of GB just for KV cache.



GPU Memory Limitations

- Even high-end GPUs like NVIDIA A100 (80GB) are insufficient.
- Weights occupy ~140GB (FP16), leaving zero room for KV cache on a single device.
- Forces multi-GPU setups or expensive offloading.



The CPU–GPU Transfer Bottleneck



Performance Mismatch

- Huge gap between processing speed and data movement speed.
- GPU Compute: >1000 TFLOPS (Tensor Cores)
- PCIe Bandwidth: Only 16–32 GB/s (Gen4/5)



vLLM's Traffic Surge

- PagedAttention enables large context but requires frequent swapping.
- Blocks are constantly moved between CPU and GPU memory.
- This heavy traffic saturates the already limited PCIe bus.

GPU Compute (TFLOPS)

PCIe Bandwidth (GB/s)

10 100 1000

Performance Scale (Logarithmic)

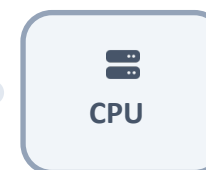
The "Compute vs. Transfer" Gap (Log Scale)

vLLM Swap Operations



High Speed Memory

PCIe Bottleneck



CPU

Slow Interconnect

Large Capacity Memory

Our Solution: Compress-Transfer-Decompress

Solving the transfer bottleneck by trading abundant compute for bandwidth

Traditional Swap

Baseline



Total Latency

3.1 ms



Compressed Swap

Proposed



Total Latency

~1.1 ms



Why cuSZp Compression Library?



GPU-Accelerated Speed

Ultra-high throughput compression specifically optimized for GPU architectures. Achieving 300-400 GB/s.



Error-Bounded Control

Strictly bounded precision loss. Guaranteed data fidelity within user-defined tolerances (e.g., 10^{-4}).



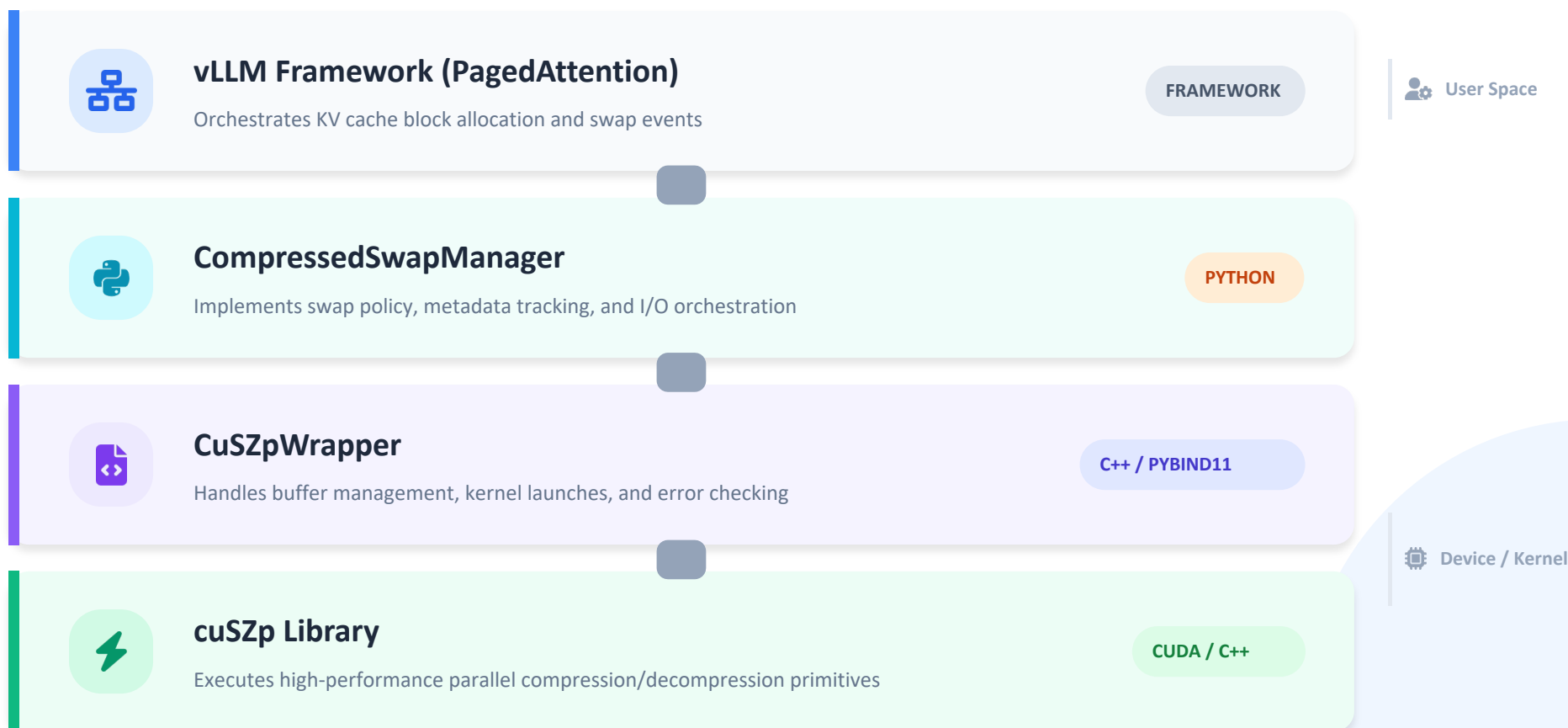
Native GPU Implementation

Entire pipeline runs on device. Zero CPU overhead and no data movement required for compression steps.

FEATURE	CUSZP	OTHER METHODS
Compression Speed	300 - 400 GB/s	Typically < 50 GB/s
Error Control	 Strict Bound	 Limited / None
GPU Native	 Yes (Full)	Often CPU-bound
Decompression	400 - 600 GB/s	Significantly Slower



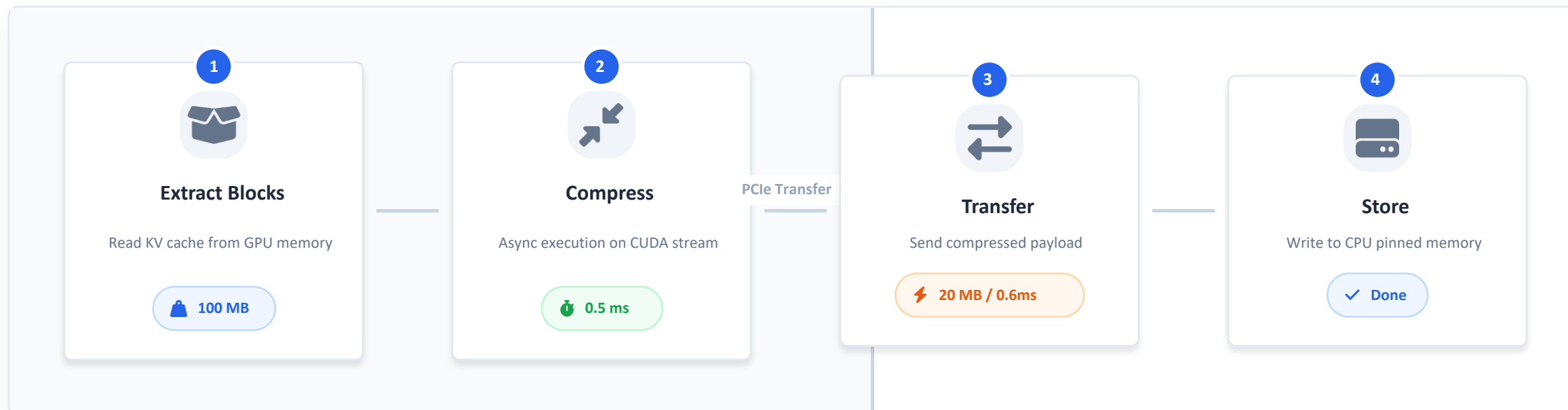
System Architecture



Compressed Swap: GPU to CPU (D2H)

🔧 DEVICE (GPU)

HOST (CPU)



TOTAL LATENCY

~1.1 ms

ORIGINAL LATENCY

3.1 ms

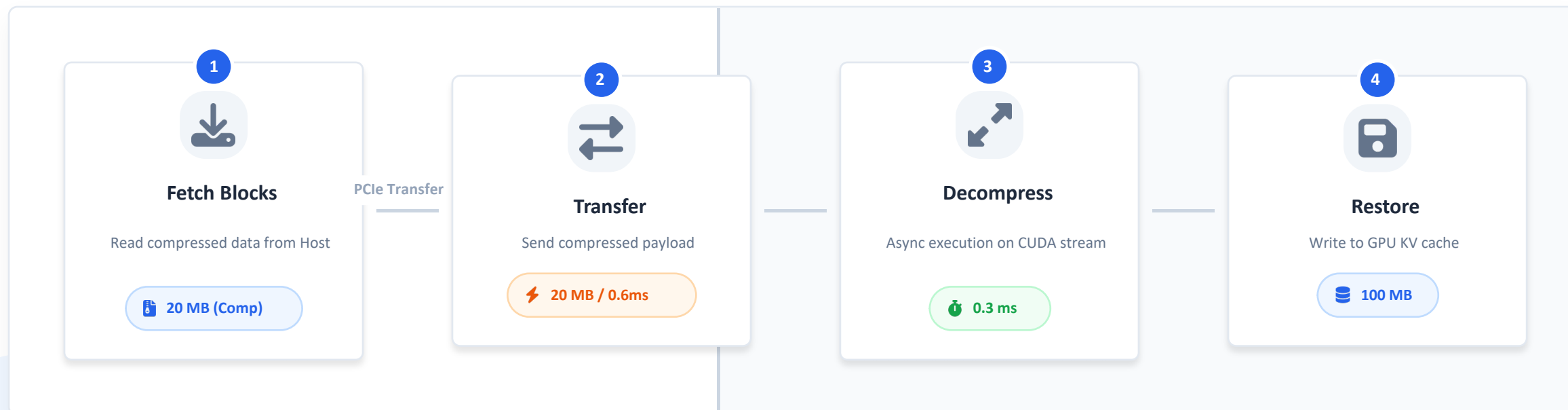
DATA REDUCTION

5.0x

Compressed Swap: CPU to GPU (H2D)

HOST (CPU)

DEVICE (GPU)



TOTAL LATENCY

~1.1 ms

ORIGINAL LATENCY

3.1 ms

PERFORMANCE GAIN

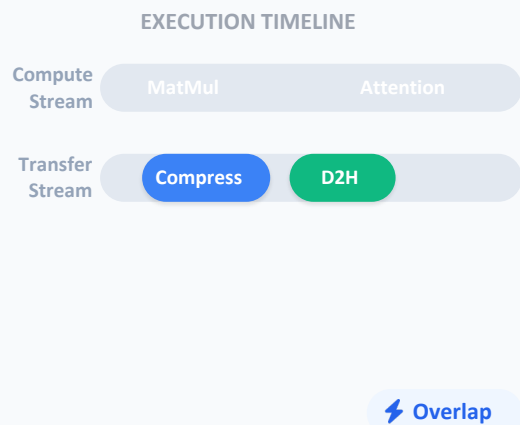
2.7x

Key Implementation Features



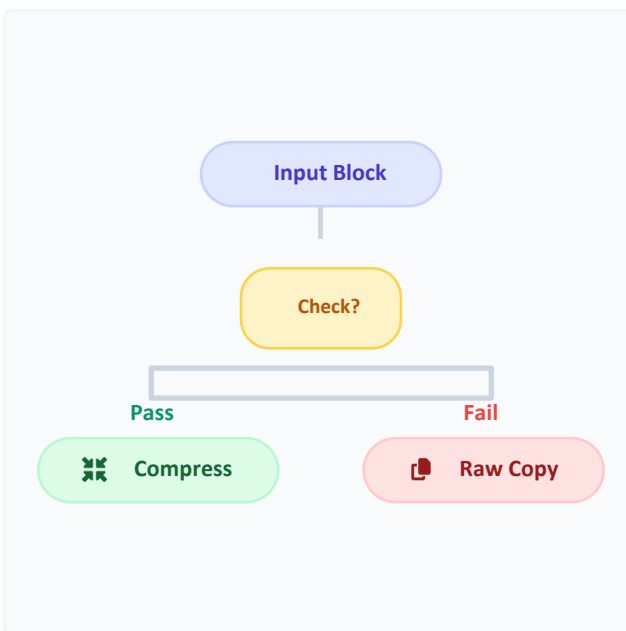
Asynchronous Execution

Leverages dedicated CUDA streams to overlap compression with computation, hiding latency.



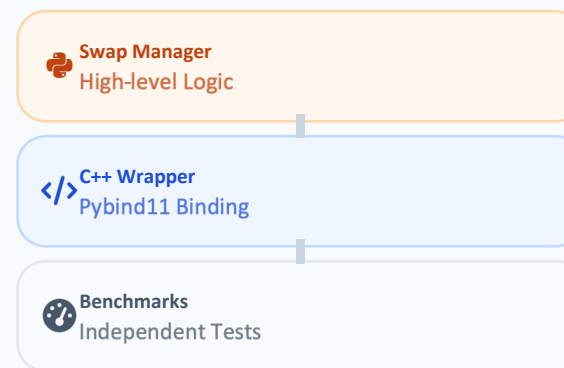
Error Handling & Fallback

Robust integrity checks ensure stability. Automatic fallback to uncompressed swap on failure.



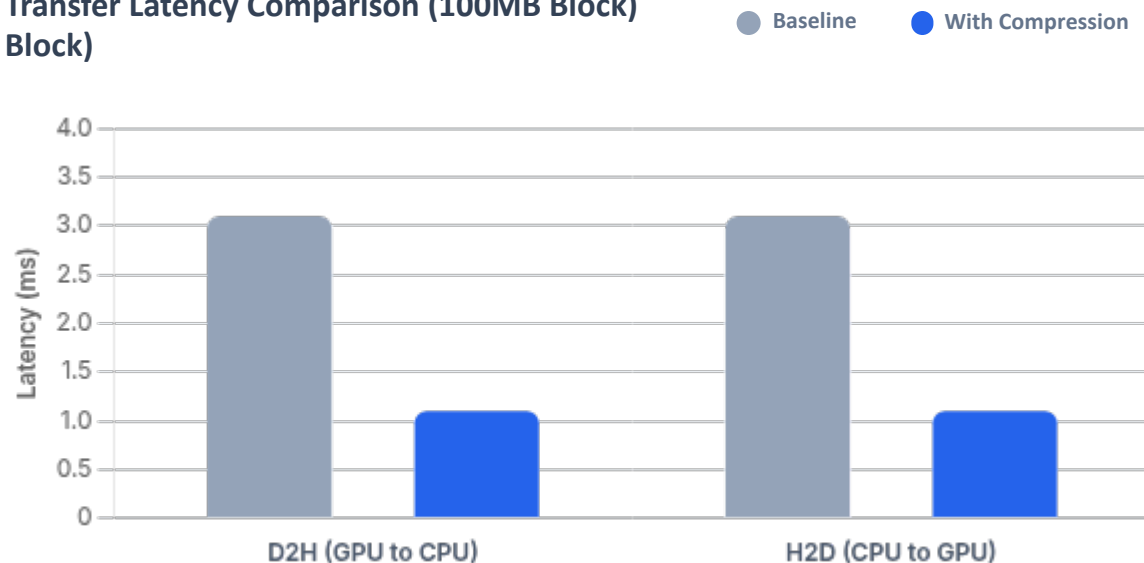
Modular Design

Clean separation of concerns facilitates maintenance and easy integration with vLLM.



Expected Performance Improvements

Transfer Latency Comparison (100MB Block)



METRIC	BASELINE	COMPRESSED
D2H Transfer	3.1 ms	1.1 ms -65%
H2D Transfer	3.1 ms	1.1 ms -65%
Compr. Speed	-	~350 GB/s
Decom. Speed	-	~500 GB/s



2.7 x

Faster Transfer



+30 %

Est. Throughput



~5 x

Compression Ratio



Why throughput improves?

By reducing the data size by 5x, we alleviate the PCIe bottleneck. The asynchronous compression hides latency behind GPU compute kernels, effectively giving us "free" bandwidth.

Accuracy Considerations & Validation

🎯 Error-Bounded Compression

cuSZp guarantees that the difference between original and decompressed values never exceeds a user-defined bound (ϵ).

i Target Tolerance: $1e-4$ (Minimal Impact)

☰ Validation Methodology



A/B Testing

Compare Logits



Perplexity Score

WikiText-2



Downstream Accuracy

MMLU / GSM8K

Compression Ratio vs. Error Bound

Finding the "Sweet Spot" for LLM Serving



* Simulated data based on typical FP16 KV cache distribution

Project Status

☰ Development Progress

~75%

✓ Completed

- ✓ Compression Pipeline Logic
- ✓ Asynchronous Execution Support
- ✓ Error Handling & Fallback Mechanism
- ✓ Benchmark Scripts
- ✓ Technical Documentation

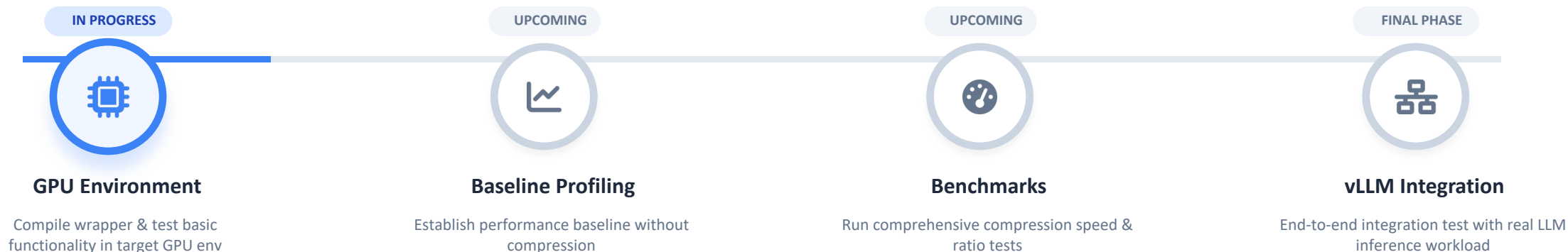
⌚ In Progress

- ⦿ GPU Environment Testing
- 📈 Performance Evaluation
- 🔍 Accuracy Validation

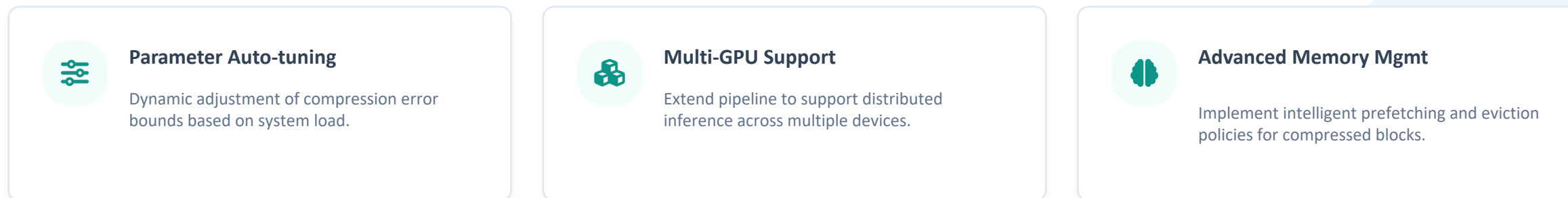
Next Milestone
End-to-End vLLM Integration

Next Steps & Future Work

IMMEDIATE NEXT STEPS



FUTURE ENHANCEMENTS



Conclusion



Key Contributions

- ✓ Developed the first GPU-native compress-transfer-decompress pipeline for LLM serving.
- ✓ Seamlessly integrated with vLLM PagedAttention memory management.
- ✓ Implemented asynchronous execution to overlap compression with computation.



Practical Value

- ↑ Mitigates the critical PCIe bottleneck in offloading scenarios.
- ✍ Incurs minimal CPU overhead, preserving host resources for other tasks.
- ✂ Scalable solution applicable to various bandwidth-constrained systems.

Thank You!

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